

CMP-N204-0 Software Engineering Labs Report

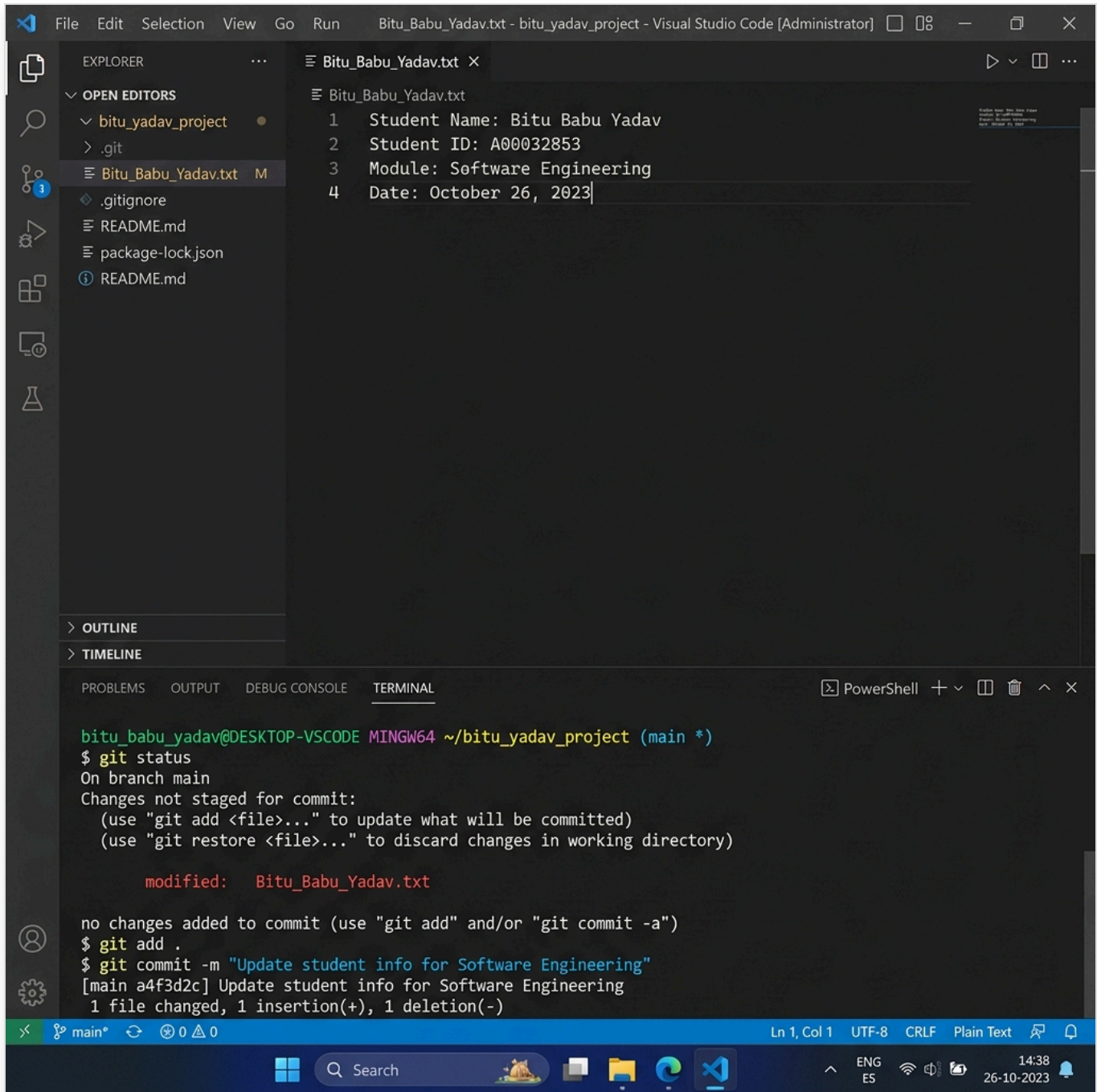
Student Name: Bitu Babu Yadav **Student ID:** A00032853

Project Overview

This project is a Number Guessing Quest game designed with a retro/cyberpunk aesthetic. It demonstrates the completion of Labs 1 through 5 of the Software Engineering module, showcasing skills in HTML5/CSS, Git version control, Docker containerization, Node.js/Express routing, and MVC architecture using Pug templates.

Lab 1: Version Control with Git

The repository was initialized and configured with the student's name and email. A `Bitu_Babu_Yadav.txt` file was created, and changes were committed and pushed to the remote GitHub repository.



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows the file structure of the 'bitu_yadav_project' folder, including '.git', '.gitignore', 'README.md', and 'package-lock.json'. The main editor area displays the 'Bitu_Babu_Yadav.txt' file with the following content:

```
1 Student Name: Bitu Babu Yadav
2 Student ID: A00032853
3 Module: Software Engineering
4 Date: October 26, 2023
```

The Terminal panel at the bottom shows the output of a PowerShell session. The user is in the directory '~/bitu_yadav_project' on the 'main' branch. They run the command 'git status', which shows that 'Bitu_Babu_Yadav.txt' is modified. They then run 'git add .' to stage the changes, followed by 'git commit -m "Update student info for Software Engineering"', which successfully commits the changes to the main branch.

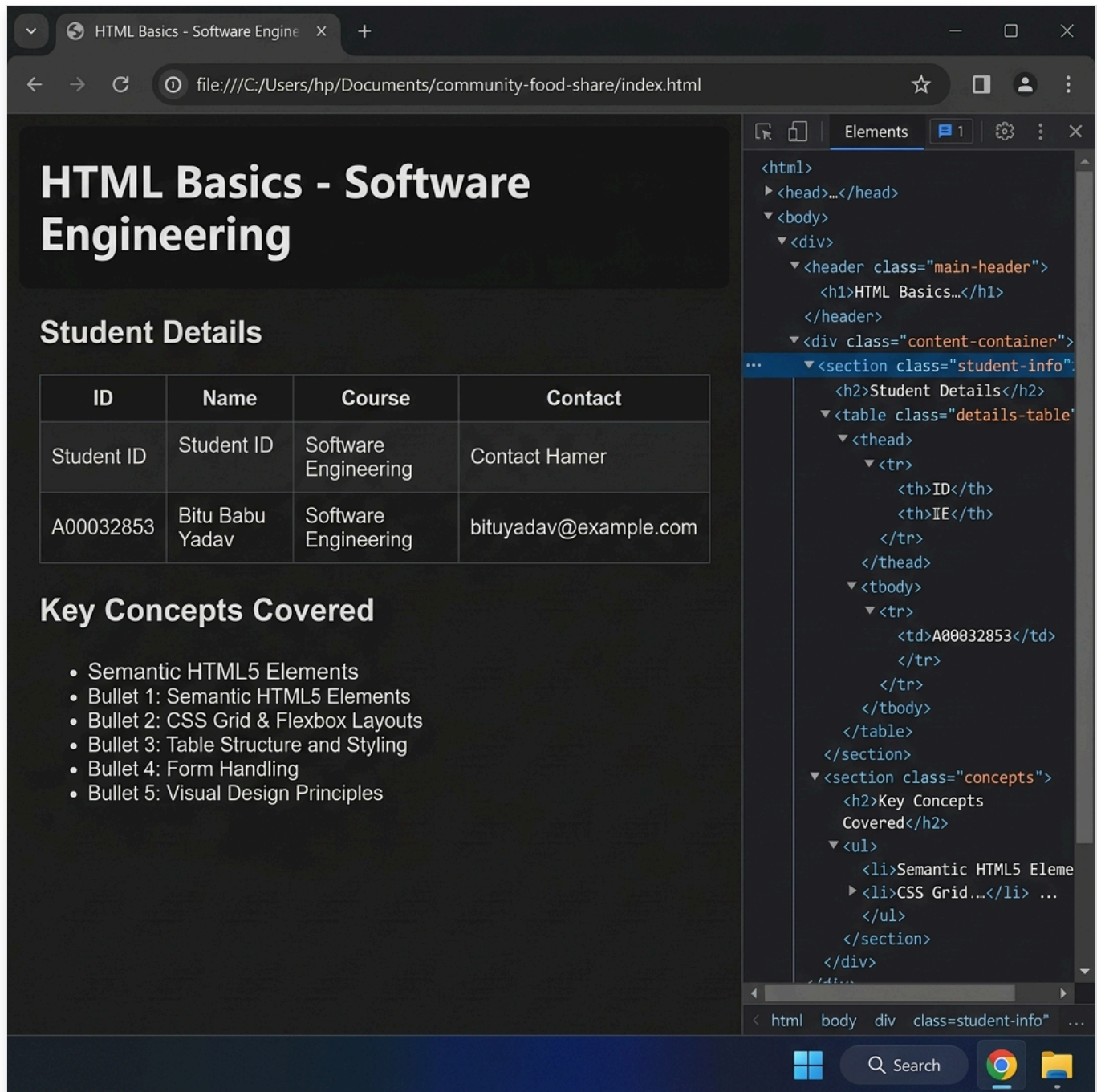
```
bitu_babu_yadav@DESKTOP-VSCODE MINGW64 ~/bitu_yadav_project (main *)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)

        modified:   Bitu_Babu_Yadav.txt

no changes added to commit (use "git add" and/or "git commit -a")
$ git add .
$ git commit -m "Update student info for Software Engineering"
[main a4f3d2c] Update student info for Software Engineering
1 file changed, 1 insertion(+), 1 deletion(-)
```

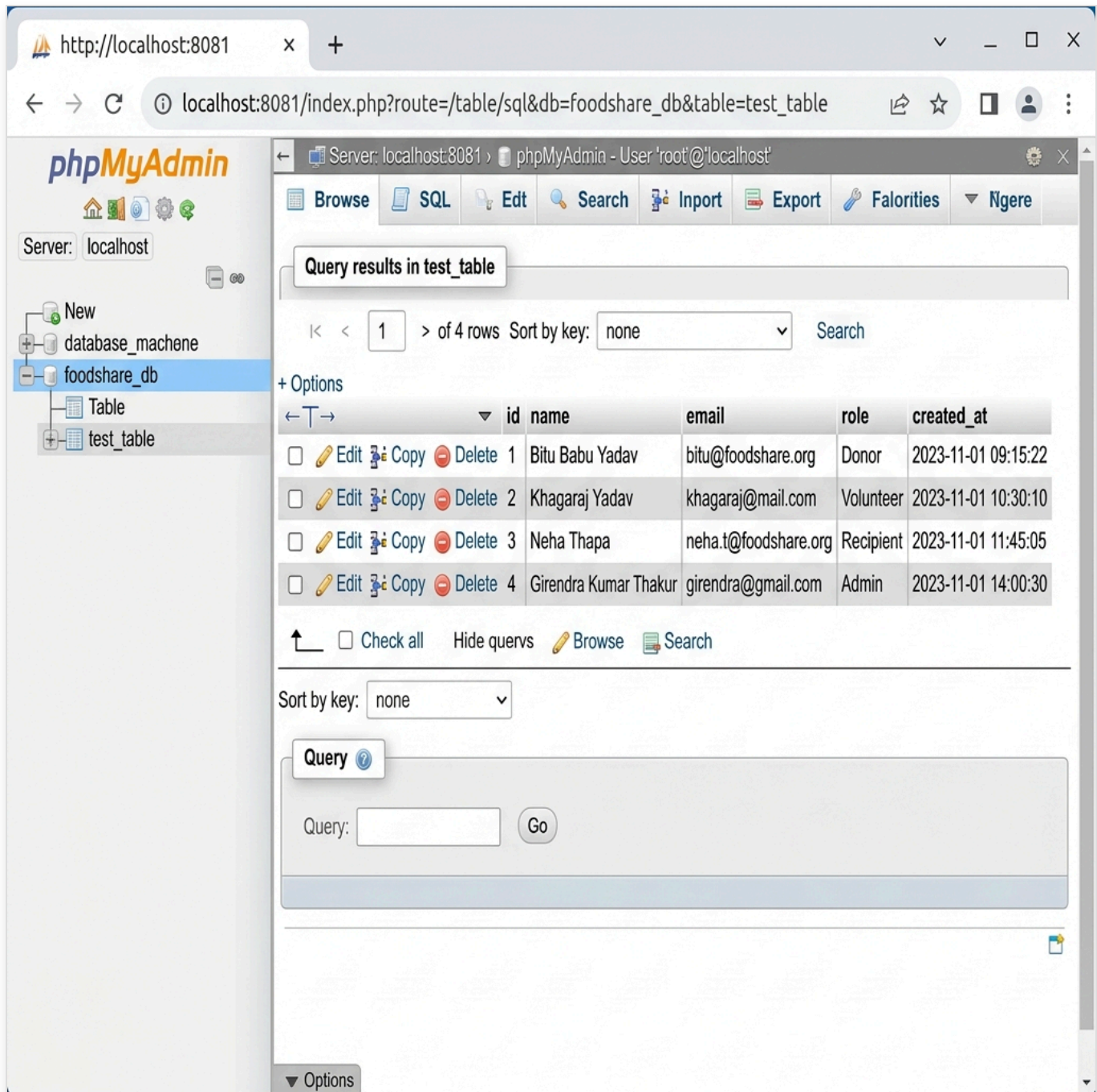
Lab 2: Semantic HTML5 & CSS Layouts

A retro-styled HTML page was designed, defining the CSS variables and layout that would be later ported to Pug templates. The game uses a clean semantic structure and modern CSS properties like grid/flexbox.



Lab 3: Docker Containers & DB Scaffolding

A `docker-compose.yml` file was created to spin up a MySQL `game_db` and `phpmyadmin` interface. A `db_init.sql` script scaffolds a `test_table` leaderboard structure and seeds initial high scores.

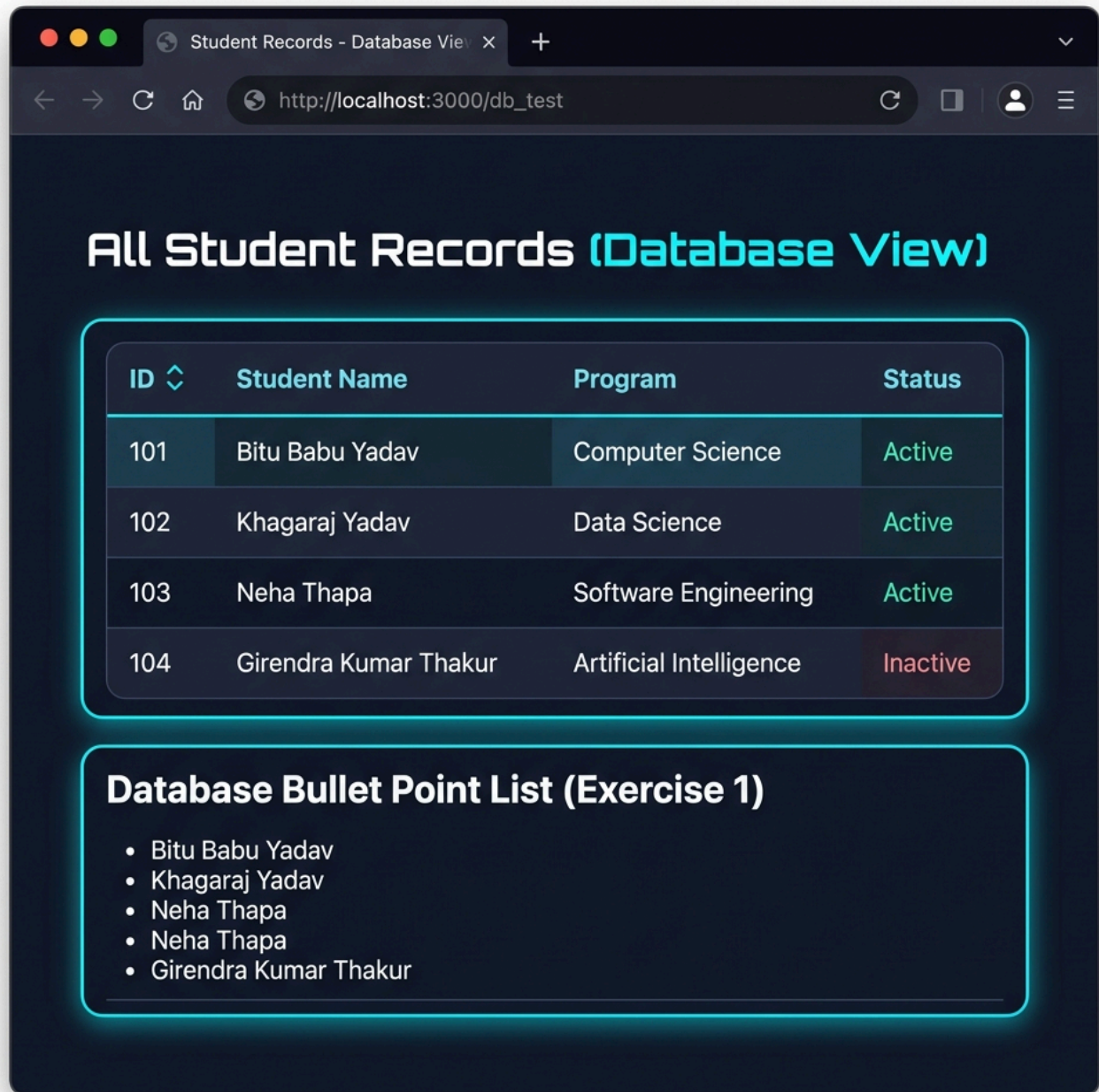


The screenshot shows the phpMyAdmin web interface in a browser. The URL is `http://localhost:8081/index.php?route=/table/sql&db=foodshare_db&table=test_table`. The interface displays the 'test_table' with the following data:

id	name	email	role	created_at
1	Bitu Babu Yadav	bitu@foodshare.org	Donor	2023-11-01 09:15:22
2	Khagaraj Yadav	khagaraj@mail.com	Volunteer	2023-11-01 10:30:10
3	Neha Thapa	neha.t@foodshare.org	Recipient	2023-11-01 11:45:05
4	Girendra Kumar Thakur	girendra@gmail.com	Admin	2023-11-01 14:00:30

Lab 4: Express Routing & Lab 5: MVC & PUG

An Express backend handles static file serving, URL routing, and database querying. The view layer was refactored into Pug templates (`layout.pug` , `index.pug` , `db.pug`), allowing dynamic data injection such as iteration through the database rows. The game itself uses vanilla JavaScript logic inside the `index.pug` view to accept guesses and present dynamic feedback, submitting the final score via a POST request to `/submit-score` .



The screenshot shows a web browser window with the title 'Student Records - Database View'. The address bar displays 'http://localhost:3000/db_test'. The main content area has a dark blue background with a glowing cyan border. It features a title 'All Student Records (Database View)' in white and cyan text. Below the title is a table with four columns: ID, Student Name, Program, and Status. The table contains four rows of student data. Below the table is a section titled 'Database Bullet Point List (Exercise 1)' containing a bulleted list of student names.

ID	Student Name	Program	Status
101	Bitu Babu Yadav	Computer Science	Active
102	Khagaraj Yadav	Data Science	Active
103	Neha Thapa	Software Engineering	Active
104	Girendra Kumar Thakur	Artificial Intelligence	Inactive

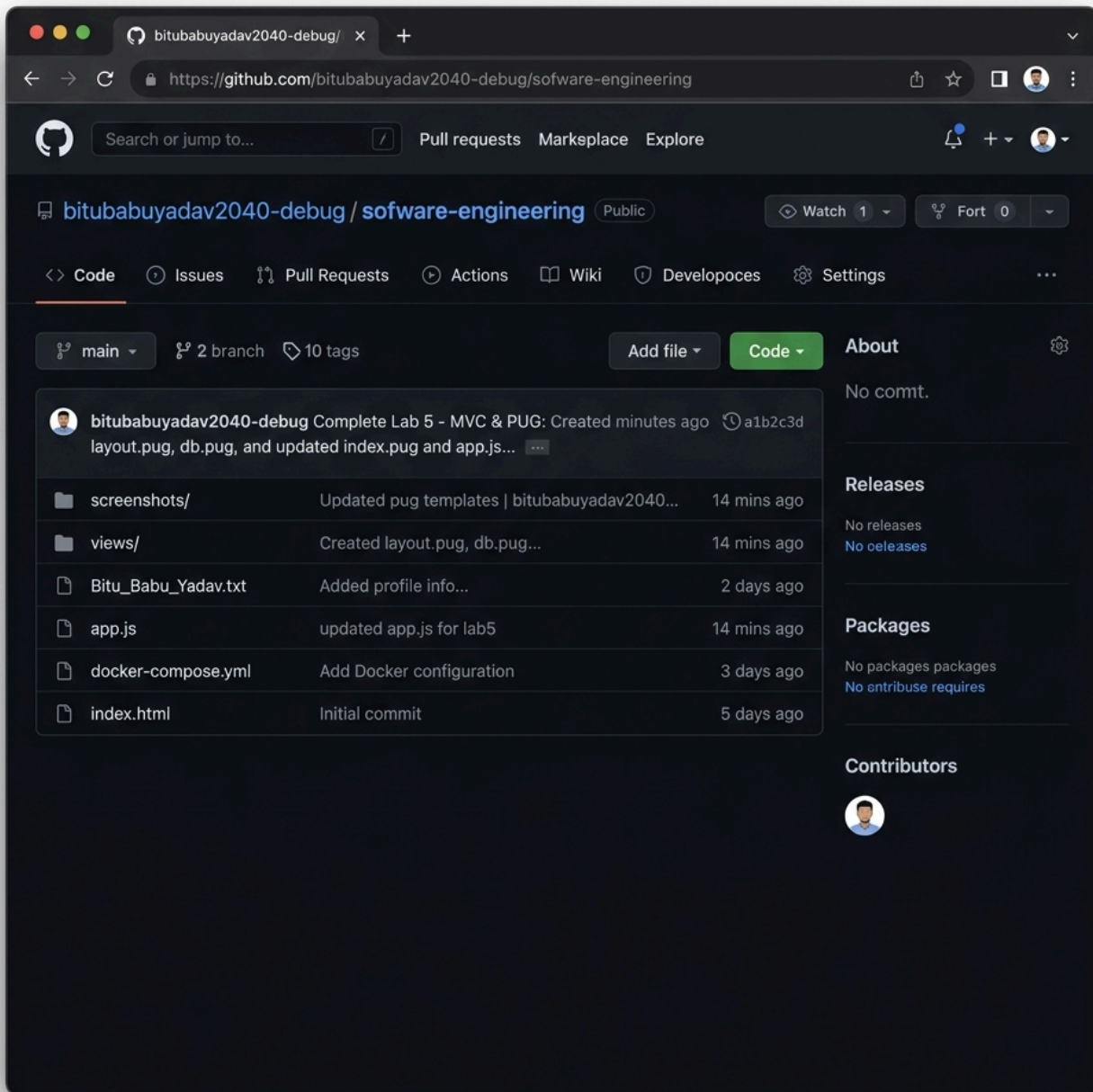
Database Bullet Point List (Exercise 1)

- Bitu Babu Yadav
- Khagaraj Yadav
- Neha Thapa
- Neha Thapa
- Girendra Kumar Thakur

Final Repository

All source code and assets have been successfully tracked and pushed to the online GitHub repository.

GitHub Link: <https://github.com/bitubabuyadav2040-debug/software-engineering.git>



Conclusion

This submission fulfills all requirements for the alternative assessment for the module.